

MDDDB-AI Hub: An Evaluation Platform for Molecular Dynamics Surrogate Models

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Molecular dynamics (MD) simulations provide detailed insights into protein dynamics, yet their data remain difficult to share and reuse. We are developing MDDDB-AI Hub, a data and model platform to support AI-driven life science research in collaboration with the European MDDDB initiative. The platform aims to offer browser-based visualization of protein dynamics for experimental researchers, standardized data deposition and sharing for computational scientists, and training datasets and pretrained MD surrogate models for AI researchers. I will introduce the motivation and design of the Hub, progress in data registration and visualization, and plans for benchmarking surrogate models under common evaluation conditions.